Validated, yet Wrong: Four Ways Small-n Metabolomics Machine Learning Passes Standard Checks

Chaiyaporn Orrudee
Artificial Intelligence Technology, School of Information Technology
King Mongkut's Institute of Technology Ladkrabang
Bangkok, Thailand
67070217@kmitl.ac.th

Kanin Calvin Shomwiwat
Artificial Intelligence Technology, School of Information Technology
King Mongkut's Institute of Technology Ladkrabang
Bangkok, Thailand
67070015@kmitl.ac.th

Abstract—Small-cohort metabolomics studies routinely report near-perfect discrimination after cross-validation, permutation testing, and feature-attribution checks. We show, on public NMR data, four failure modes in which every one of these standard checks passes while the reported result is wrong or uninterpretable: (i) an outcome manufactured from the features through change scores, (ii) exogenous contaminants that feature-removal faithfulness tests rank as highly as genuine biology, (iii) a perfect acquisition-batch/label confound, and (iv) a label whose name does not match what it encodes. For each, we give the standard check it passes and the targeted control that exposes it. In a bariatric-surgery responder forecast, a label-permutation test gave $p = 0.002$, yet a pairing-destroyed null scored higher than the observed model, and after correcting the outcome the trained model (PR-AUC 0.483) did no better than a learn-nothing baseline (0.522). In simulation with no forecastable individual response, label permutation rejected in 11–100% of datasets at $\alpha = 0.05$, rising with sample size; residualising change on each marker's own baseline still rejected in 8–29%, whereas residualising on all baselines jointly held the rate at 5–8%. We release the controls as a reusable checklist with reference code.

Index Terms—metabolomics, NMR, validation, data leakage, regression to the mean, permutation test, feature attribution, batch effects, reproducibility

I. Introduction

Machine-learning studies on small metabolomics cohorts ($n < 200$) frequently report AUC above 0.95. Readers are asked to trust these numbers because the authors used cross-validation, a permutation test, and an explanation method such as SHAP [12]. Yet leakage and related evaluation errors have been documented across hundreds of ML-based studies in many fields [8], and omics data add hazards of their own: acquisition batch, sample handling, and repeated measurements of the same person [9]. With a few dozen subjects, any of these can produce a near-perfect score.

Each of these checks answers a narrow question. Cross-validation asks whether the model generalises to new samples drawn the same way; a label permutation asks whether the features carry any association with the label; a faithfulness test asks whether the model relies on a feature. None asks whether the label measures what it claims to, or whether the association is biological. A study can therefore pass all three checks and still be wrong.

This paper makes four contributions:

1) A taxonomy of four failure modes, each paired with the standard check it passes and a targeted control that catches it (Table I).
2) A pairing-destroyed null for change-score outcomes, shown to detect a manufactured outcome that label permutation accepts.
3) A simulation study (36,000 datasets) showing that label permutation becomes more wrong as n grows, that the common per-marker regression-to-the-mean correction is itself liberal for correlated markers, and that joint residualisation restores near-nominal error.
4) An open checklist and reference implementations of the controls for metabolomics ML studies.

II. Related Work

A. Regression to the mean and mathematical coupling

Correlating change with initial value is a known statistical artefact [5]–[7], and a change score that shares terms with the baseline is mathematically coupled to it [15]: measurement error and within-person variation make extreme baselines move back toward the mean, so baseline "predicts" change even when nothing happens. The epidemiological remedies (analysis of covariance, Oldham's

Code, fixed partitions, and verification scripts: [TODO: repository URL and commit hash].

method) are well known [16], [17]. Our contribution is not the artefact but its interaction with ML practice: an outcome defined from change scores, a flexible model given the baselines as features, and a label permutation as the significance test. We show that this combination certifies the artefact, and that the per-marker correction a careful analyst might add is not sufficient.

B. Leakage and confounding in ML-based science

Kapoor and Narayanan [8] catalogue leakage types, including features that act as proxies for the outcome and non-independence between training and test data. Batch effects can be perfectly confounded with the biological variable of interest, in which case no analysis can separate them [9]; analytical bias in serum metabolomics is well documented [18], [19]. Reporting guidelines such as TRIPOD+AI [10] require that the outcome definition and its relation to the predictors be stated. Our cases are concrete metabolomics instances of these categories, each paired with a test that detects it.

C. Feature-attribution faithfulness

SHAP [12] attributes a prediction to features; ROAR [11] tests whether the attributed features matter by removing them and retraining. Both answer a question about the model. Attribution methods can mislead even about model behaviour [20], [21]. A feature can be faithfully important to a model and still be an artefact of how the samples were collected. We use a known exogenous contaminant as a positive control to show that ROAR cannot tell the two apart.

D. Permutation tests for classifiers

A label permutation tests the null hypothesis that features and labels are independent [13]. Rejecting it shows only that some association exists. If the label is computed from the features, that association is guaranteed, and the test answers a question nobody asked. The permutation must also refit the model on every permuted label vector: permuting labels against fixed out-of-fold scores is not a valid test (Section VIII); the broader fragility of resampling-based validation on small cohorts is well established [24]–[27].

III. Data and Common Setup

A. Datasets

MTBLS242 [1], [2]: serum ^{1}H-NMR from a bariatric surgery cohort; 106 preoperative and 71 twelve-month samples (465 samples over five time points); 21 quantified metabolites. Patient identifiers are recoverable from sample names with the pattern `^(\d+)[-_](\d+)[-_]S$`.

MTBLS1 [3]: urine ^{1}H-NMR, type 2 diabetes vs. control; 132 samples from 40–42 subjects (12 volunteers sampled at seven time points; 28–30 patients whose sample-to-patient membership is not deposited; see Section VI).

MTBLS881 [28]: serum ^{1}H-NMR from 50 adults undergoing cardiac surgery with cardiopulmonary bypass, sampled at six times (from −1 to 20 h) from two sites (left atrium, LA; pulmonary artery, PA). We use it only to replicate Case 1 on a second cohort: 47 patients have both the −1 h and 20 h samples at each site.

B. Models and evaluation

We deliberately reuse the models from the original analyses rather than tuning new ones, since the point is to evaluate the checks, not the models. Classifiers are XGBoost [14] with monotonic constraints whose directions were assigned from the metabolomics literature (Cases 2–4) and L2-penalised logistic regression (Case 1). The headline metric is PR-AUC (average precision), reported next to the prevalence it must beat. Cross-validation is repeated and grouped by subject wherever subjects can be identified, and intervals come from bootstrap resampling of subjects. Every analysis includes a learn-nothing baseline: a fixed, unfitted score that any trained model must beat.

IV. Case 1: An Outcome Manufactured from the Features

A. Setup

Task: predict from the preoperative profile whether a patient is a twelve-month "strong responder," defined as $\geq$9 of 11 markers changing in the expected direction. 71 paired patients, 31 responders.

B. Why the standard check passes

Baseline x_0 is both the predictor and a term in the outcome $\Delta = x_{12} - x_0$. Under pure noise, $\mathrm{corr}(x_0, \Delta) < 0$ by construction; the observed median was -0.62. A label permutation breaks the label–feature link entirely, so it tests against a null that the coupled outcome trivially beats: $p = 0.002$.

C. Exposing control: pairing-destroyed null

Shuffle which twelve-month sample belongs to which subject, recompute the outcome from scratch, and refit. This keeps the arithmetic coupling and removes any real individual-level signal. Because shuffling breaks the positive correlation between visits, the null has stronger coupling than the observed data; the test is therefore conservative (Section VIII) and is best read as a diagnostic: a model that does not beat it has shown nothing beyond regression to the mean.

D. Result

Original design (raw change, $\geq$9 of 11 markers, contaminants included). The trained model reaches PR-AUC 0.683 against a prevalence of 0.437 (the originally reported 0.671 does not reproduce exactly on re-run, as the repository's own correction notes). A label permutation with the model refitted on every permutation gives $p = 0.001$ (0.002 with the original fixed-score procedure). The pairing-destroyed null has median lift 2.04× and the observed lift is 1.57×, below 499 of 500 null replicates (Fig. 1). The model

TABLE I
Four failure modes: the standard check each passes and the control that exposes it.

Case	Failure mode	Data	Reported result	Standard check passed	Exposing control
1	Outcome manufactured from features	MTBLS242	PR-AUC 0.683	Label permutation, $p = 0.002$	Pairing-destroyed null
2	Contaminant ranked as biology	MTBLS242	PR-AUC 0.972	ROAR faithfulness	Known-artefact positive control
3	Batch equals label	MTBLS1	PR-AUC 0.976	Subject-grouped CV	Metadata/batch audit
4	Label name ≠ label meaning	MTBLS242	ROC-AUC 0.973	Subject-grouped CV	Label-provenance audit

TABLE II
Case 1, corrected designs. Gap = model − learn-nothing scorer PR-AUC; interval from a bootstrap over the 71 subjects. Prevalence differs because the joint design yields a different responder count, so compare lifts and gaps, not PR-AUC levels.

	Per-marker resid.	Joint resid.
Responders / 71	31	37
Prevalence	0.437	0.521
Model PR-AUC	0.483	0.554
Learn-nothing PR-AUC	0.522	0.589
Gap	−0.042	−0.037
95% interval	[−0.149, +0.052]	[−0.111, +0.037]
Refit permutation p	1.000	1.000
Pairing-null lift, median	1.04	1.05

cannot be distinguished from regression to the mean. In the original solver the largest coefficient was 0.018, and an unfitted scorer matched its PR-AUC to three decimals (0.653).

Corrected design (change residualised on baseline inside each training fold, contaminants removed). Two variants are shown in Table II. In both, the trained model does not beat the learn-nothing scorer, and refitted permutation tests give no evidence of signal ($p = 1.0$). The repeat-level bootstrap interval for the gap excludes zero on the wrong side, but it understates uncertainty by treating repeats as independent; resampling the 71 subjects gives an interval that spans zero. The defensible statement is that the model is no better than the scorer, not that it is worse.

E. Replication on a second cohort

We repeated the four tests on MTBLS881 with the design fixed in a pre-registration file before any model was fitted: baseline −1 h versus 20 h; both sites analysed separately (LA declared primary); 11 marker metabolites chosen in advance; improvement direction taken from training-fold mean change only; K chosen so prevalence is near 0.5; exogenous drugs (propofol, heparin, ethanol) excluded. The model is unconstrained ridge logistic regression with the same repeated CV as Case 1. This is a different, acute setting (hours after surgery, not a year), so it tests the statistical artefact, not the surgical biology. With 47 patients every interval is wide (Table III).

Raw change replicates. At both sites a label permutation rejects ($p = 0.007$), yet the pairing-destroyed null reaches

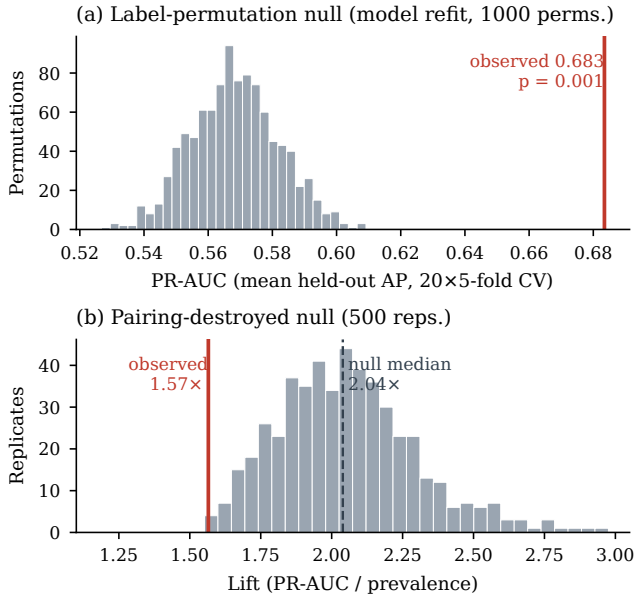


Fig. 1. Original (withdrawn) design. (a) Label-permutation null of average precision, model refitted per permutation. (b) Pairing-destroyed null of lift. The label permutation rejects; the pairing-destroyed null sits above the observed model.

equal or higher lift (median 1.49 and 1.40 vs. observed 1.29 and 1.19), and the model does not beat the learn-nothing scorer (gap −0.037 and +0.013, both intervals spanning zero). Refitting the full model on each shuffle puts the observed lift inside the null ($p = 0.86$ at LA, 0.91 at PA; 150 replicates).

Residualised outcomes replicate only partly. At PA, joint residualisation shows no signal ($p = 0.977$), matching MTBLS242. At LA the per-marker outcome is permutation-significant ($p = 0.003$), and for LA joint residualisation the permutation test is not significant ($p = 0.103$) while the model's lift (1.45) exceeds the scorer-based pairing-null 95th percentile (1.27). The tests disagree here, and 47 patients cannot resolve it. Every gap interval except one (below) includes zero.

Exogenous compounds create apparent signal. Adding propofol, heparin and ethanol to the features and markers made the joint-residualised outcome permutation-significant at both sites ($p = 0.005$, the floor for 200

permutations), against $p = 0.103$ and 0.977 without them; at LA the gap became +0.163 [+0.069, +0.252]. We did not test the mechanism, but the pattern matches Case 2: drugs administered during surgery distinguish samples because of handling, not metabolism.

V. Case 2: Contaminants Ranked as Biology

A. Setup

ROAR [11] removes the top-k attributed features, retrains, and compares the score with removing k random features. Here the attribution is SHAP on the MTBLS242 pre- vs. 12-month task (177 samples, 21 metabolites). We repeat the original analysis with three fixes: cross-validation is grouped by patient (10 repeats of 5 folds), attribution is recomputed after every retrain (recursive ROAR), and the random null uses 500 distinct draws per k (exhaustive for $k \leq 2$) under the same folds as the informed runs. Grouped baseline PR-AUC is 0.973; ungrouped CV gives 0.972.

B. Why the standard check passes

Recursive ROAR removes, in order, dimethyl sulfone, isopropanol, L-valine, glycine, methanol, and then further metabolites. From $k = 3$ onward the curve lies below almost the entire random band (null percentile 0.0–0.2), so ROAR reports that the removed features genuinely matter to the model. That is true. Dimethyl sulfone, isopropanol and methanol are exogenous, and isopropanol is a skin-preparation agent used in surgery. They distinguish pre- from post-operative samples because of how the samples were handled, not because of metabolism.

C. Exposing control: artefact vs. biology

At $k = 3$, removing the contaminant trio gives PR-AUC 0.954 (null percentile 0.2); removing three biological markers (lipoproteins, L-valine, L-tyrosine) gives 0.959 (percentile 1.2). The 0.006 gap is far smaller than the fold standard deviation of about 0.03. ROAR cannot separate the two.

A spiked positive control makes the point without relying on this cohort. We add a synthetic feature $d\,y + \mathcal{N}(0, 1)$, which depends only on the label and carries no biology, and run recursive ROAR (10 draws per d, Table IV). At $d = 1.75$ the spike is ranked first by both SHAP and ROAR, and removing it costs more than removing any real metabolite (0.0127 vs. 0.0073 for dimethyl sulfone). ROAR ranks a feature by how much signal it carries, whatever its origin.

D. Caveats

This is one cohort, and its label is confounded with surgery (Case 4). Random draws overlap and share folds, so percentiles below the band mean "outside the random range," not calibrated p-values. Attribution used all 177 samples, which makes informed removal slightly optimistic. The multi-seed ranking places L-valine above lipoproteins, the reverse of the original ranking; we kept the original biology trio for comparability.

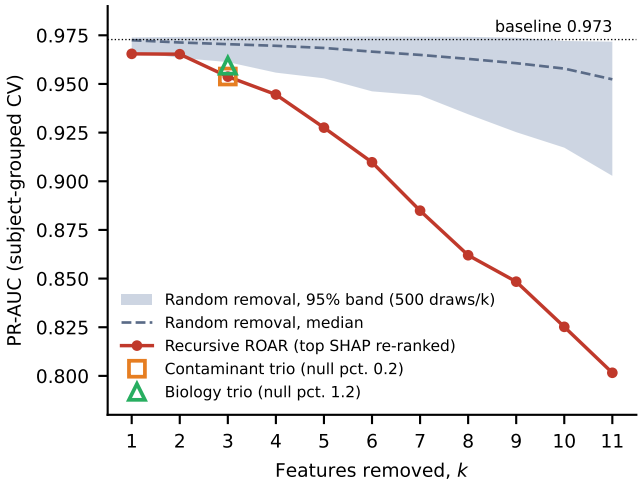


Fig. 2. Recursive ROAR on MTBLS242 against 500-draw random-removal bands. The informed curve lies below the band from $k = 3$ onward whether or not the removed features are biological.

VI. Case 3: Batch Equals Label

A. Setup

The Case 3 model is the original monotonic XGBoost on 132 urine spectra, evaluated by CV grouped by subject. With seed 42 it reproduces the reported PR-AUC of 0.976; over 20 CV seeds the mean is 0.911 (range 0.845–0.975), so the headline was a favourable split.

B. Why the standard check passes

Subject-grouped CV prevents the same person from appearing in training and test folds. It cannot detect a confound shared by every member of a class.

C. Exposing control: metadata audit

We cross-tabulated the label against all 64 columns of the deposited sample and assay sheets. Of these, 56 are constant (instrument, probe, pulse sequence, temperature) and five are per-sample identifiers. Only sex varies and is not an identifier. Every control sample name begins with ADG19007u and every diabetic sample name with ADG10003u (Table V). A logistic model given only this prefix, under the same grouped CV, scores PR-AUC 1.000 and ROC-AUC 1.000 against a prevalence of 0.364. No within-batch comparison is possible because each batch contains one class. The deposited protocol also gives the two groups different age and BMI criteria and different sampling schedules (seven time points for volunteers vs. one to three for patients), which are therefore aliased with the label as well.

Sex is imbalanced (controls 66.7% male vs. cases 45.8%) but not confounded. Subject identity is not deposited: the control numbering yields exactly 12 volunteers, matching

TABLE III
Case 1 replication on MTBLS881 ($n = 47$ per site). Gap = model − learn-nothing scorer PR-AUC, subject-bootstrap interval. K = number of the 11 markers required to define a responder.

Site	Outcome	Prev.	PR-AUC	Scorer PR-AUC	Gap [95% CI]	p_{perm} (refit)	Lift	Pair-null lift med. (p95)
LA	Raw change (K=10)	0.49	0.632	0.695	−0.037 [−0.231, +0.118]	0.007	1.29	1.49 (2.09)
LA	Per-marker resid. (K=6)	0.55	0.678	0.542	+0.164 [−0.004, +0.309]	0.003	1.23	1.05 (1.25)
LA	Joint resid. (K=6)	0.43	0.619	0.545	+0.071 [−0.077, +0.190]	0.103	1.45	1.06 (1.27)
PA	Raw change (K=10)	0.64	0.757	0.723	+0.013 [−0.111, +0.154]	0.007	1.19	1.40 (1.89)
PA	Per-marker resid. (K=6)	0.51	0.631	0.546	+0.078 [−0.086, +0.222]	0.096	1.24	1.05 (1.27)
PA	Joint resid. (K=6)	0.53	0.598	0.629	−0.034 [−0.169, +0.092]	0.977	1.12	1.06 (1.23)

TABLE IV
Spiked artefact: median rank among 22 features (1 = most important).

d	SHAP rank	ROAR rank
0.75	10	14.5
1.25	3	3
1.75	1	1

TABLE V
MTBLS1 samples by label, sample-name prefix and sex. The prefix determines the label exactly; sex does not.

Label	Prefix	F	M	Samples	Subjects
Control	ADG19007u	28	56	84	12
T2DM	ADG10003u	26	22	48	28–30
Total		54	78	132	40–42

the protocol, while the 48 diabetic samples come from 28–30 patients whose sample-to-patient membership cannot be recovered. A numbering heuristic used in our original analysis merged at least three pairs of distinct patients. Merging errs toward conservative CV, and ungrouped CV (0.969) differs little, so grouping does not drive the result; batch does, and it cannot be removed.

VII. Case 4: The Label Is Not What It Is Called

The "obese vs. non-obese" task on MTBLS242 is in fact "preoperative vs. twelve months after surgery": 71 of 106 patients appear in both classes. Subject-grouped CV (0.969) and naive CV (0.970) agree, so the issue is not leakage but label meaning. A classifier trained on this label learns everything that differs between a patient before surgery and the same patient a year later: weight loss, but also post-surgical diet, supplements and medication, fasting state, and the year-long gap in sample storage. Its score is not evidence that the serum profile identifies obesity in a general population, where lean and obese people differ in age, sex and lifestyle rather than in surgical history. The correct description is "pre- vs. post-bariatric-surgery classification," and the check that exposes the mismatch is not statistical: write down, in one sentence, what physically distinguishes the two classes.

VIII. Simulation Study

A. Design

Each subject i has a persistent level μ_{ij} for $p = 11$ outcome markers, with a shared factor (inter-marker correlation $\rho = 0.3$), and ten auxiliary features $z_i \sim \mathcal{N}(0, I)$. Observed visits are

$$x_{ij}^{(0)} = \sqrt{r}\,\mu_{ij} + \sqrt{1-r}\,\varepsilon_{ij}^{(0)}, \tag{1}$$

$$x_{ij}^{(12)} = \sqrt{r}\,(\mu_{ij} + \gamma\, g_i\, s_j) + \sqrt{1-r}\,\varepsilon_{ij}^{(12)}, \tag{2}$$

where r is the reliability (share of variance that persists between visits), $s_j \in \{\pm 1\}$ the expected direction of marker j, and $g_i = w^\top z_i / \|w\|$ a forecastable response propensity. With $\gamma = 0$ no individual response exists, so any association between baseline and outcome is regression to the mean. The outcome is the MTBLS242-style rule "$\geq$6 of 11 markers change in the expected direction." The model is ridge regression ($\lambda = 10$) on the 21 standardised baseline features, scored out-of-fold in 5-fold CV by average precision.

We vary $n \in \{50, 100, 200, 500\}$, $r \in \{0.25, 0.5, 0.75\}$ and $\gamma \in \{0, 0.25, 0.5\}$, with 1,000 datasets per cell. Setting $r = 0.25$ reproduces the median baseline–change correlation of -0.62 observed in MTBLS242. Each dataset is tested four ways at $\alpha = 0.05$, each with 200 null replicates and the model refitted on every replicate:

1) Label permutation on the raw-change outcome.
2) Pairing-destroyed null on lift (AP over prevalence).
3) Per-marker residualisation: within each training fold, regress change$_j$ on baseline$_j$; label from residuals; permute.
4) Joint residualisation: as (3), but regress change$_j$ on all 11 baselines.

A permutation test that holds out-of-fold scores fixed and permutes only the labels rejected 9.7% of pure-noise datasets in our sanity check; refitting on each permutation [13] gave 5.3%, so all tests refit.

B. Results

Table VI and Fig. 3 give the rejection rates.

Label permutation is wrong, and more so with more data. With no individual response, it rejected in 26% of datasets at $n = 50$ and 100% at $n = 500$ when $r =$

TABLE VI
False-positive rate (%) with no individual response ($\gamma = 0$), $\alpha = 0.05$, 1,000 datasets per row. Wilson 95% half-width $\leq$3.1 points.

r	n	Label perm.	Pair-null	Per-marker	Joint
0.25	50	26.1	4.3	8.0	7.4
0.25	100	52.0	2.1	9.3	6.2
0.25	200	92.6	0.7	12.0	5.9
0.25	500	100.0	0.3	18.9	6.3
0.50	50	16.3	2.2	10.9	8.1
0.50	100	35.9	0.6	13.1	6.9
0.50	200	74.4	0.2	16.3	6.8
0.50	500	99.6	0.0	28.8	6.7
0.75	50	10.6	0.9	9.6	6.4
0.75	100	18.8	0.2	13.3	6.7
0.75	200	40.4	0.0	16.2	6.5
0.75	500	90.3	0.0	27.0	5.3

0.25; even at $r = 0.75$ it reached 90% at $n = 500$. More subjects estimate the regression-to-the-mean artefact more precisely; they do not remove it.

Per-marker residualisation is not enough. Residualising each marker's change on its own baseline left false-positive rates of 8–29%, again rising with n. When markers are correlated, residual change in marker j remains correlated with the baselines of the other markers, which the model also sees.

Joint residualisation is near nominal. Regressing each change on all baselines kept the rate at 5.3–8.1%, slightly liberal at $n = 50$, while retaining power similar to label permutation when a real response exists (e.g. 97.5% vs. 100% at $\gamma = 0.25$, $r = 0.25$, $n = 500$).

The pairing-destroyed null is valid but conservative. Its false-positive rate never exceeded 4.3%, but at the MTBLS242-like reliability $r = 0.25$ and $\gamma = 0.25$ it had no power at any n (4–5%). It detects manufactured outcomes; it should not be the only test of a real one.

C. Implication for Case 1

The per-marker correction used in the original analysis is liberal, so Section IV repeats the corrected analysis with joint residualisation. The conclusion is unchanged: the model does no better than an unfitted scorer ($p = 1.0$ with refitting). A liberal test that finds nothing, followed by a better-calibrated test that also finds nothing, is consistent evidence of absence at this sample size.

IX. Discussion

A. A checklist for small-n metabolomics ML

1) If the outcome is computed from the features, do not rely on a label permutation. Residualise change on all baselines inside each training fold, and report a pairing-destroyed null as a diagnostic.
2) Refit the model inside every permutation replicate.
3) Include a learn-nothing baseline and report the gap with an interval.
4) Audit features for exogenous compounds; report results with and without them.
5) Cross-tabulate the label against every metadata column (batch, run order, sample-name prefix, sex).
6) State in one sentence what the label physically encodes.

B. Limitations

The empirical cases use three small public datasets, chosen because their problems are verifiable from the published files. The findings concern validation practice; they say nothing for or against the biology of obesity or diabetes. Regression to the mean is classical statistics, and our contribution is limited to showing how ML practice fails to detect it. The simulation uses Gaussian data, a linear ridge model and one outcome rule; non-linear models and other rules may differ in degree. The pairing-destroyed null is conservative and can miss real effects at low reliability, so it should be a diagnostic, not the sole test. Finally, the joint residualisation removes any true biological dependence of change on baseline together with the artefact; separating the two needs repeated baseline measurements.

X. Conclusion

Cross-validation, permutation tests and attribution checks are necessary but answer narrow questions. In four public-data cases, each passed while the result was wrong: an outcome built from its own features, contaminants ranked as biology, a batch equal to the label, and a label named for something it did not measure. Each failure has a cheap, targeted control. We recommend that small-n metabolomics ML studies report these controls alongside their headline score, and we release code that runs them.

Data and Code Availability

MTBLS242, MTBLS1 and MTBLS881 are public through MetaboLights [4]. [TODO: Repository URL, commit, test-suite instructions.]

Acknowledgment

[TODO: Funding, advisors.]

Use of Generative AI

The authors used Claude (Anthropic) to assist with analysis code, the simulation, and drafting text. The authors reviewed all code and text, re-ran every reported analysis, and take full responsibility for the content. [TODO: Adjust wording to the target venue's policy.]

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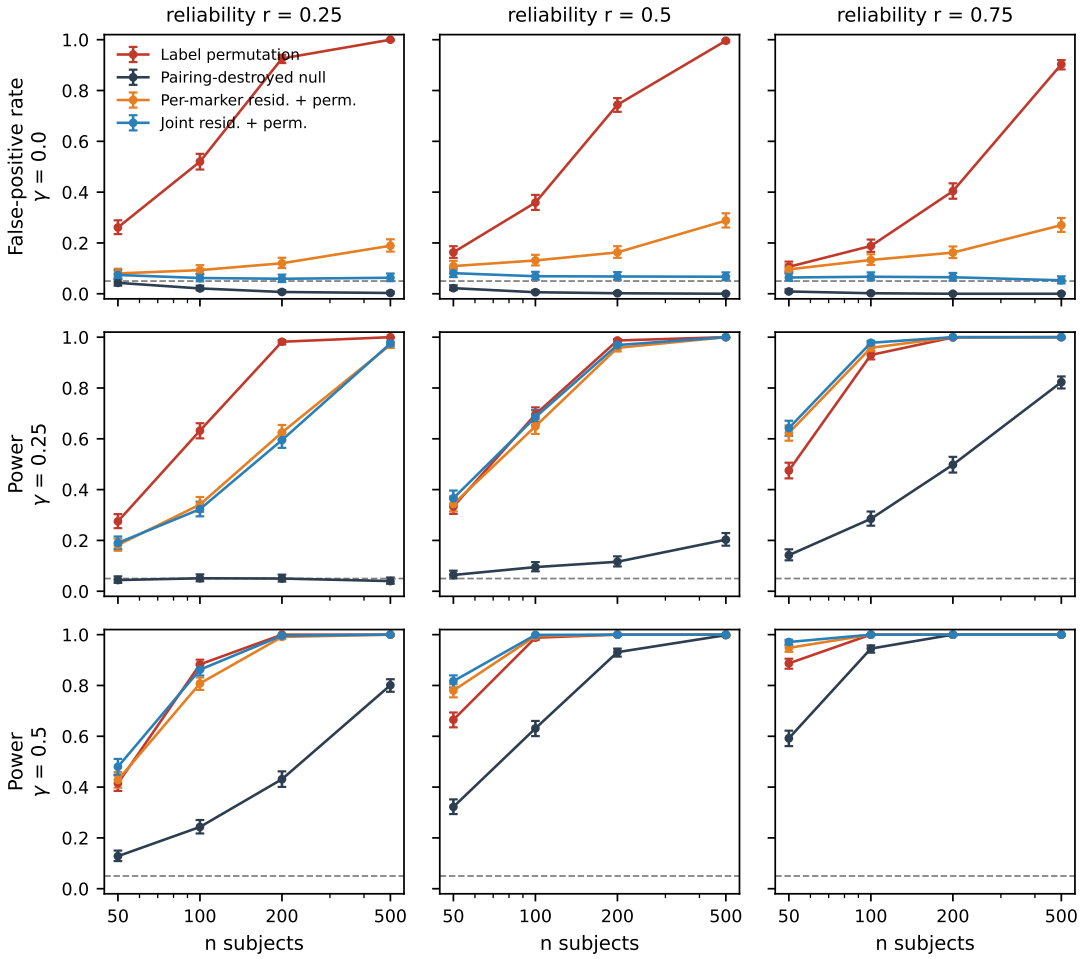


Fig. 3. Rejection rate vs. number of subjects. Top row: false-positive rate with no individual response ($\gamma = 0$). Lower rows: power with a real, forecastable response. Columns: reliability r. Dashed line: $\alpha = 0.05$. Error bars: Wilson 95% intervals over 1,000 datasets.

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